Shaping the land

Earth's surface changes constantly but so gradually we can hardly see it. Wind, waves, moving ice, and other forces wear away rocks and mountains and create valleys. At the same time, Earth's plates move, forming mountains and continents.

EROSION

Water, wind, and ice wear down rocks and soil. They also move the resulting materials to new places, and in doing so change the shape of the land. The process is called erosion. Natural forces cause most erosion but human activity, such as deforestation, also contributes.

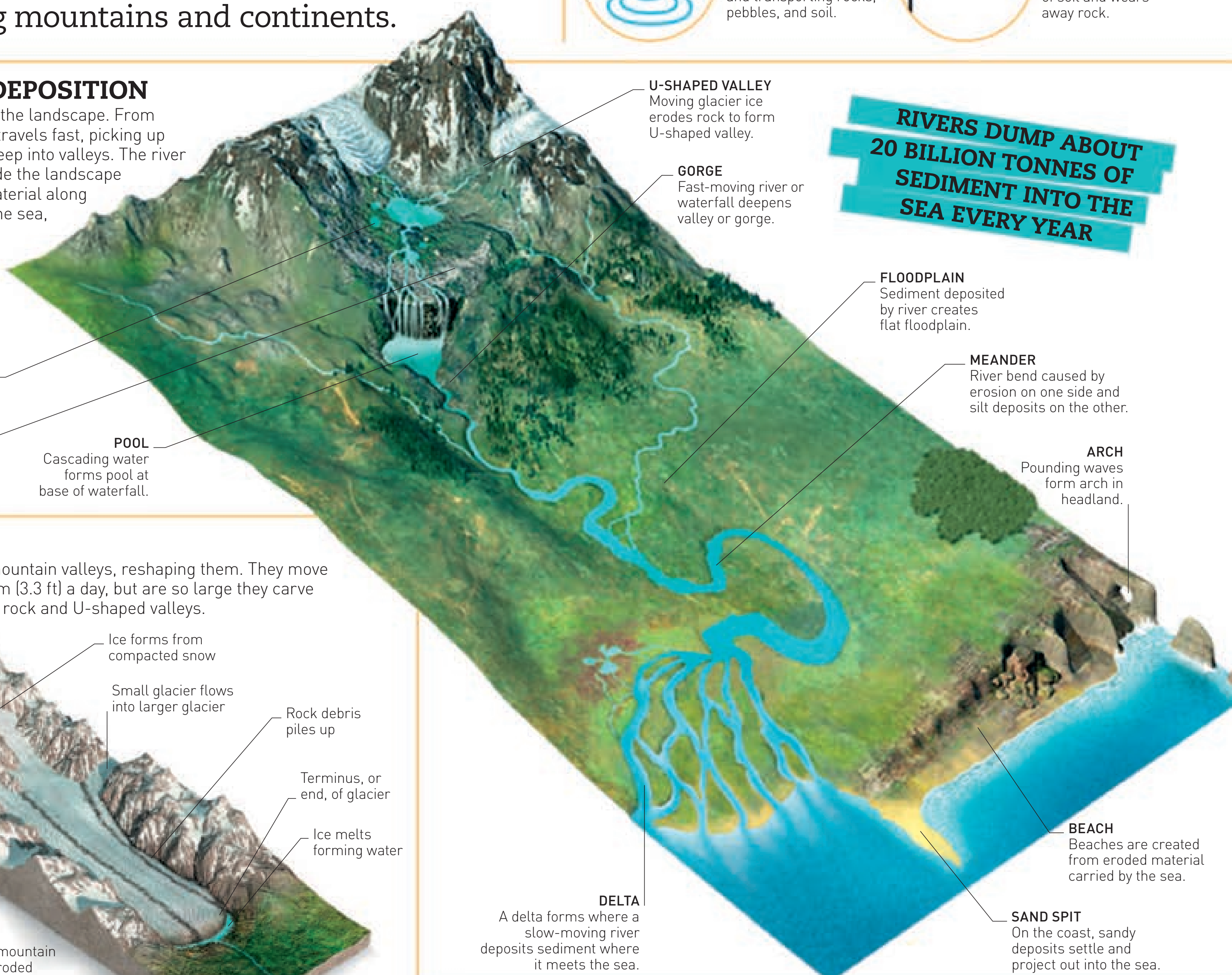
GLACIER
Huge ice masses called glaciers scrape away rocks and earth as they move down mountain valleys.

WATER
Moving water erodes coasts, cliffs, and riverbanks, picking up and transporting rocks, pebbles, and soil.

WIND
A powerful erosive force, wind blows away the top surface of soil and wears away rock.

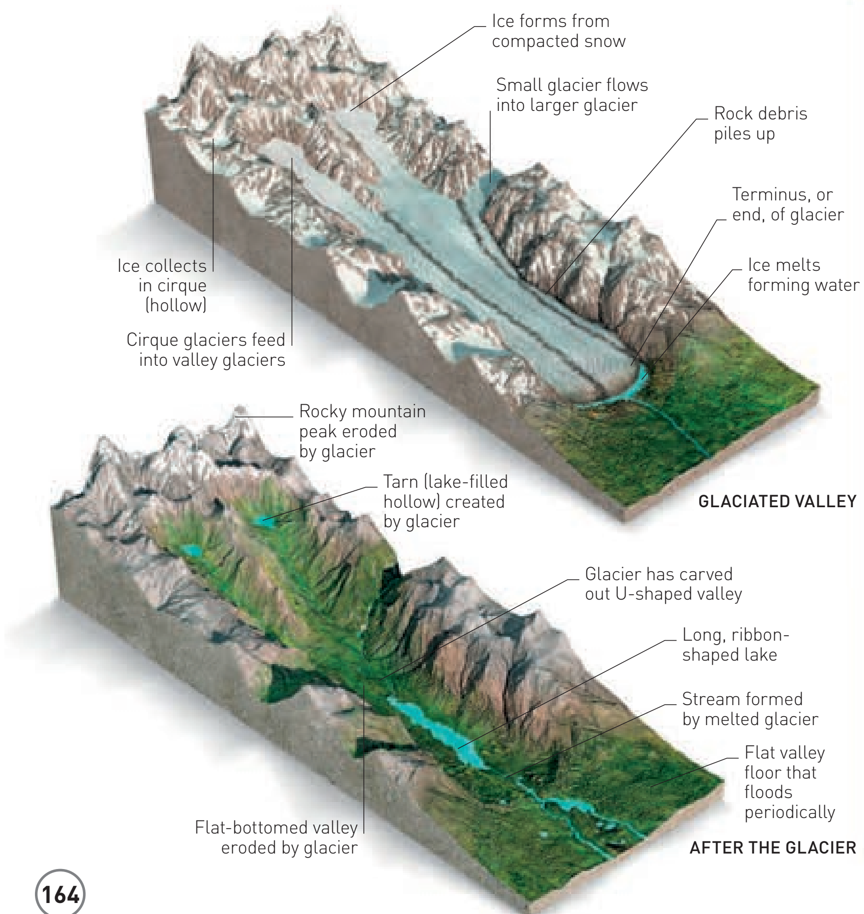
EROSION AND DEPOSITION

Rivers and streams mould the landscape. From glacier beginnings, a river travels fast, picking up rocky debris and carving deep into valleys. The river slows but continues to erode the landscape and also deposits some material along the way. When it reaches the sea, it drops silt to form deltas and beaches.



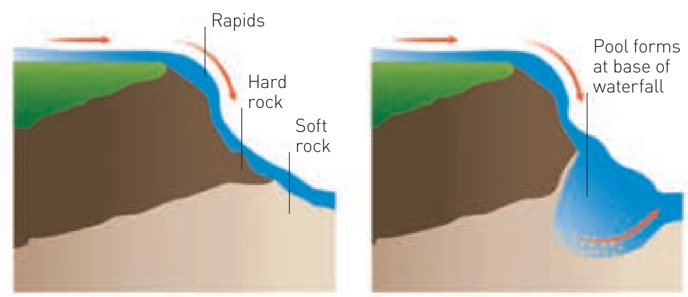
GLACIERS

Icy glaciers flow through mountain valleys, reshaping them. They move slowly, usually less than 1 m (3.3 ft) a day, but are so large they carve out vast depressions in the rock and U-shaped valleys.



WATERFALL FORMATION

A waterfall forms when a river pours over a rocky edge. The water flow erodes the rock, creating a pool and undermining the ledge. Soft rock erodes more quickly than hard rock so the amount of erosion varies, as does the height and flow of the water.



RAPIDS
Rapids occur when the flow of a shallow river is broken up by hard rock projecting out of the water.

WATERFALL
When a river erodes soft rock, beyond the rapids, it carves out a pool into which the water cascades.